

Planning

[Requirements] Elicitation

John Ceroli, jceroli@oakhill.nsw.edu.au High School Teacher

1. How often do you use quiz tools like Kahoot in your classroom or learning sessions?

"I try to use interactive quizzes once or twice a week, mainly at the end of key lessons to review key concepts."

2. What challenges or frustrations do you face while using quiz tools for learning or teaching?

"The main challenge I face when using these types of quizzes is students become easily distracted due to the competitiveness of the game. They have more of a focus towards the game aspect rather than learning. Sometimes the time limit can be too short for students to read and answer the questions properly, especially for more complex topics."

3. Are there any specific features or functionalities you feel are missing in tools like Kahoot that would improve your experience?

"My current quiz platform locks all students to the same questions, however I would love to see a feature where I can tailor question difficulty for different students. Some students need more time or simpler questions, whereas other more advanced students could benefit from more challenging content. Also, I would be really eager to see a method to track student progress over time as opposed to just one quiz."

4. Do you find quiz tools like Kahoot effective for engaging students or learners, or do you believe there's a better way to engage them?

"Quizzes platforms like Kahoot are effective to an extent, specifically for reviewing basic knowledge. The interactive element really engages students, however I perceive my quiz platform to be more of an engagement tool rather than a tool that drives deep learning. I like to combine this with more detailed assessments or discussions after the quiz to really reinforce the concepts."

Xianing Berroeta Tay, xianingb@gmail.com, University Student

1. How often do you use quiz tools like Kahoot in your classroom or learning sessions?

“Quite often. Lecturers often use kahoot-like quizzes at the beginning of lectures to recap on the content from the week before. So maybe 2-3 times per week.”

2. What challenges or frustrations do you face while using quiz tools for learning or teaching?

“ I think that as a tool for memorisation and quick retrieval of information it serves a good purpose. However, I find that it doesn't give good feedback on where a student is at with their learning. It's frustrating that once I have completed a quiz, I have no tangible results to indicate my level of understanding. “

3. Are there any specific features or functionalities you feel are missing in tools like Kahoot that would improve your experience?

“I think if there were an option to display analytics at the end of a quiz to show my rank within the quiz or what percentile I scored compared to my peers, it would improve the use of a quiz platform like kahoot.

4. Do you find quiz tools like Kahoot effective for engaging students or learners, or do you believe there's a better way to engage them?

“I find quizzes like kahoot quite engaging initially for students due to the competition it creates, but after sometime I have realised the learning experience is not optimal. As I said before, oftentimes there is no metric of performance to reflect on and gauge understanding.”

[Requirements] Analysis & Specification - User stories and Use cases

User Story 1:

- **As a High School Teacher, I want the ability to** customise quiz difficulty levels tailored to the diverse academic levels within my classrooms, **so that** I can better accommodate the learning needs of my students.

Acceptance Criteria:

- System allows users to create quizzes with varying difficulty levels.
- The quiz host (teacher) can assign specific difficulty levels to a specific user (student).
- The system ensures only questions from a specified difficulty level are assigned to a student.
- The quiz host has the option to change the difficulty level for a specific user at any time.
- Students are notified of their difficulty level prior to commencing the quiz.

User Story 2:

- **As a High School Teacher, I want the ability to** track my students' progress overtime both collectively and individually, **so that** I can monitor areas for improvement and provide additional support whereby necessary

Acceptance Criteria:

- The system stores a record of each student's scores, and attributes like time taken, and number of attempts, under 'Class Performance' tab on the menu bar.
- Once the tab is interacted with, quiz host (teacher) can view individual student performances both graphically and tabularly.
- The system holds some analytical capacity, generating performance trends for each student over time, highlighting areas of concern.
- Quiz hosts can filter data based on specific criteria (content areas, time period, student, difficulty level, etc.)

User Story 3:

- **As a University Student, I want the ability to** receive immediate feedback on my progress in the quizzes and an evaluation of my performance, **so that** I can use quizzes to test my understanding and knowledge in a topic.

Acceptance Criteria:

- After answers are revealed for a question in a quiz, for a participant a quiz ranking as well as a graphic showing a plot of quiz points is displayed.
 - What percentile a participant finishes in is calculated.
 - At the end of the quiz a final ranking and percentile is displayed on the device of the quiz participant.
 - A 'find out more' button is displayed, next to final ranking and percentile numbers.
 - When the 'find out more' button is pressed, past results in the same quiz are displayed with graphs, given that a participant has played this quiz before.
 - Quiz participants can remain on this page even if the quiz host ends the quiz session.
 - Quiz participants can access this data at any time through a 'past results' page.
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Use Cases:

User Story 1:

Use Case: Customising Quiz Difficulty Levels

Goal in Context: The teacher should be able to customise difficulty levels of quizzes for individual students, allowing them to tailor to each student's academic level. This ensures that regardless of ability, all students can effectively engage with the content.

Scope: This is applicable to using quiz platforms in a High School setting, where quizzes can be tailored to varying student ability.

Preconditions:

- Teacher is logged onto the quiz system.
- Quiz has already been created by the teacher.
- The system supports defining and assigning multiple difficulty levels for quiz questions.
- Students are registered and assigned to their respective classes.

Success End Condition:

- The quiz is deployed with each student receiving difficulty-tailored questions.
- Students are notified of their difficulty level prior to commencing the quiz.
- The teacher can modify the quiz's difficulty level for individual students at any given time.

Failed End Condition:

- System incorrectly assigns specified difficulty levels to students, or students are not made aware of their difficulty level.
- Teacher is unable to modify the student's difficulty level throughout the duration of the quiz.
- Students receive questions beyond their difficulty scope.

Primary Actor: Teacher (Quiz Host)

Trigger: The teacher initiates the quiz creation process and begins the assignment of difficulty level to students.

User Story 3:

Use Case: Providing Quiz Participant Analytics

Goal in Context: Quiz Participants should receive detailed feedback on their performance in any given quiz. This allows students a way to use quizzes to gauge academic performance and understanding, in the same way an exam does.

Scope: For use in educational institutions, such as high schools and universities, where students want to receive feedback on their learning to achieve their goals.

Preconditions:

- Quiz platform which has a quiz host and quiz participants.
- The quiz host displays questions from their electronic device.
- The quiz participants answer questions from their electronic device.
- Quiz system has a points system which gives participants points based on the accuracy and speed of their answers.
- Quiz system returns to participants device points gained after each question round.

Success End Condition:

- Quiz feedback implemented such that quiz participant's ranking and percentile score is displayed to them on their device at the end of a quiz round.
- Quiz feedback implemented such that quiz participant's ranking and graphic of quiz points is displayed to them on their device after each question.
- Quiz feedback has a 'find out more' button, which allows users to display a graphic and compare with previous attempts.
- Quiz feedback is accessible after the quiz is ended and at any time through the 'past results' section.

Failed End Condition:

- Quiz feedback is not displayed for quiz participants at all throughout the quiz.
- Quiz feedback for all participants is displayed on the quiz host's device rather than a participant's individual device.
- Quiz feedback is implemented such that ranking and percentile score are calculated incorrectly.
- Quiz feedback is unable to be reassessed after quiz has ended
- Current Quiz feedback is unable to be compared with past quiz feedback.

Primary Actor: Student (Quiz Participant)

Trigger: Student joins quiz which has been started by a quiz host.

[Requirements] Validation

John Ceroli:

Use Case 1:

- “Yes this is ideally the change I’m looking forward to. One additional feature I’d like to see is automatic feedback on incorrect answers, so that when a student selects an incorrect answer, they should get a detailed explanation on why the correct answer was what it was. Whether this be through creator input or generative AI like ChatGPT. This will really assist in reinforcing learning immediately.”
- Suggestion: integrate immediate feedback following an incorrect answer.

Xianing Berroeta Tay:

Use Case 3:

- “The implementation of Quiz feedback during and after the quiz describes the solution I was searching for. I think this type of system would really improve my experience using quizzes as a learning tool rather than just a fun game. Thinking of another implementation which could be an addition to this use case, I would suggest a tagging system whereby a quiz can tag questions to a certain topic, and the person doing the quiz can see from which topics they are most often getting problems incorrect. ”
- Suggestion/Improvement: Implement feature in which quiz questions are ‘tagged’ and extra feedback is relayed about which type of questions they are getting most often incorrect.